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Supplementary appendix

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HRQoL and pain outcomes in the VISION study of ^{177}Lu-PSMA-617

Supplementary appendix

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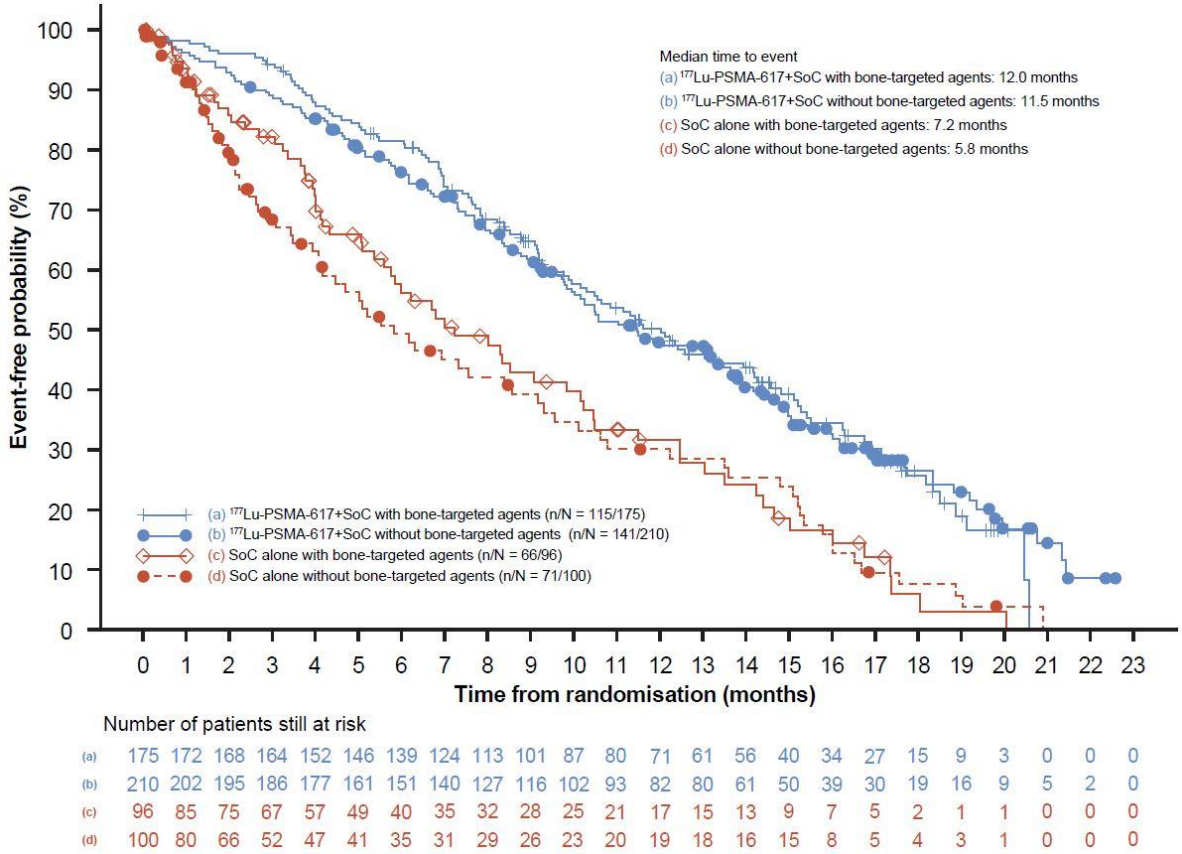
Figure S1: Time to first SSE or death by concurrent use of bone-targeted agents as part of standard of care



Number of patients still at risk

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
(a)	175	172	168	164	152	146	139	124	113	101	87	80	71	61	56	40	34	27	15	9	3	0	0	0
(b)	210	202	195	186	177	161	151	140	127	116	102	93	82	80	61	50	39	30	19	16	9	5	2	0
(c)	96	85	75	67	57	49	40	35	32	28	25	21	17	15	13	9	7	5	2	1	1	0	0	0
(d)	100	80	66	52	47	41	35	31	29	26	23	20	19	18	16	15	8	5	4	3	1	0	0	0

n=total number of events. N=total number of patients. PSMA=prostate-specific membrane antigen. SoC=standard of care. SSE=symptomatic skeletal event. Testing was two-sided using the Cox model (Wald Chi-square test). Analyses in the 581 patients randomised after measures were implemented on or after March 5, 2019

Table S1: Characteristics of the patients at baseline

	^{177}Lu-PSMA-617 plus SoC (n = 385)	SoC alone (n = 196)
Median age, year (IQR)	71·0 (65–75)	72·0 (66–76)
Race, n (%)		
White	336 (87.3)	166 (84.7)
Black or African American	29 (7.5)	14 (7.1)
Asian	6 (1.6)	9 (4.6)
Other[a]	2 (0.5)	0
Missing	12 (3.1)	7 (3.6)
ECOG performance-status score of 0 or 1, n (%)[b]	352 (91)	179 (91)
Site of disease, n (%)		
Lung	35 (9)	20 (10)
Liver	47 (12)	26 (13)
Lymph node	193 (50)	99 (51)
Bone	351 (91)	179 (91)
Median PSA level, ng/mL (IQR)	93·2 (29·0–293·0)	90·7 (29·2–323·2)
Median alkaline phosphatase level, IU/L (IQR)	108·0 (70·0–170·0)	96·0 (70·0–158·0)
Median LDH, IU/L (IQR)[c]	230·5 (192·5–315·0)	232·0 (190·0–291·0)
Median time since diagnosis, year (IQR)	7·3 (3·78–11·43)	7·0 (4·14–12·91)
Gleason score at diagnosis, n (%)[d]		
8–10	226 (59)	118 (60)
Unknown	28 (7)	19 (10)
Previous prostatectomy, n (%)[e]	159 (41)	82 (42)
Previous ARPI, n (%)[f]		
One regimen	213 (55)	98 (50)
Two regimens	150 (39)	86 (44)
More than two regimens	22 (6)	12 (6)
Previous taxane therapy, n (%)[g]		
One regimen	207 (54)	102 (52)
Two regimens	173 (45)	92 (47)
Docetaxel	377 (98)	191 (97)
Cabazitaxel	161 (42)	84 (43)

ARPI=androgen receptor pathway inhibitor. ECOG=Eastern Cooperative Oncology Group. IQR=interquartile range. LDH=lactate dehydrogenase. PSA=prostate-specific antigen. SoC=standard of care.

[a]Other includes Native Hawaiian or Other Pacific Islander, American Indian or Alaska Native and more than one race reported.

[b]ECOG Performance Status scores range from 0 to 5, with higher numbers indicating greater disability.

[c]Data were missing for a very small number of patients (*Table* S3).

[d]Scores on the Gleason scale range from 2 to 10, with higher scores indicating a worse prognosis. A score of 8 to 10 indicates high-grade cancer. In the remaining patients whose score was known, the Gleason score was 2 to 7 (intermediate or low-grade cancer).

[e]Data exclude biopsy and include prostatectomy, radical prostatectomy, transurethral prostatectomy, cystoprostatectomy, and retropubic prostatectomy.

[f]ARPIs were defined as enzalutamide, abiraterone, and apalutamide.

[g]Taxanes were defined as cabazitaxel, docetaxel, and paclitaxel. Of the 831 patients, 8 (1%) had received more than two taxanes previously. The use of taxanes was well balanced between treatment groups in both analysis sets.

Analyses in the 581 patients randomised after measures were implemented on or after March 5, 2019.

Table S2: Patient-reported outcome questionnaire completion rates

	FACT-P		BPI-SF		EQ-5D-5L	
	^{177}Lu-PSMA-617 plus SoC (n = 385) n (%)	**SoC alone (n = 196) n (%)**	**^{177}Lu-PSMA-617 plus SoC (n = 385) n (%)**	**SoC alone (n = 196) n (%)**	**^{177}Lu-PSMA-617 plus SoC (n = 385) n (%)**	**SoC alone (n = 196) n (%)**
Baseline						
Number of patients on study	385	196	385	196	385	196
Completed	367 (95)	186 (95)	369 (96)	186 (95)	367 (95)	186 (95)
by patient	269 (70)	140 (71)	271 (70)	140 (71)	269 (70)	140 (71)
with assistance of caretaker	20 (5)	7 (4)	20 (5)	7 (4)	20 (5)	7 (4)
with assistance of clinician	78 (20)	39 (20)	78 (20)	39 (20)	78 (20)	39 (20)
Cycle 1, week 1, day 1						
Number of patients on study	369	164	369	164	369	164
Completed	349 (95)	146 (89)	357 (97)	151 (92)	349 (95)	147 (90)
by patient	244 (66)	114 (70)	249 (67)	115 (70)	244 (66)	114 (70)
with assistance of caretaker	17 (5)	4 (2)	16 (4)	5 (3)	16 (4)	4 (2)
with assistance of clinician	88 (24)	28 (17)	92 (25)	31 (19)	89 (24)	29 (18)
Cycle 2, week 1, day 1						
Number of patients on study	349	115	349	115	349	115
Completed	335 (96)	102 (89)	339 (97)	102 (89)	335 (96)	102 (89)
by patient	237 (68)	69 (60)	239 (68)	69 (60)	237 (68)	69 (60)
with assistance of caretaker	9 (3)	6 (5)	10 (3)	6 (5)	9 (3)	6 (5)
with assistance of clinician	89 (26)	27 (23)	90 (26)	27 (23)	89 (26)	27 (23)
Cycle 3, week 1, day 1						
Number of patients on study	308	61	308	61	308	61
Completed	292 (95)	55 (90)	297 (96)	54 (89)	296 (96)	55 (90)
with assistance of caretaker	198 (64)	37 (61)	202 (66)	36 (59)	202 (66)	37 (61)
with assistance of clinician	15 (5)	0	16 (5)	0	15 (5)	0
by clinician	79 (26)	18 (30)	79 (26)	18 (30)	79 (26)	18 (30)
Cycle 4, week 1, day 1						
Number of patients on study	252	36	252	36	252	36
Completed	243 (96)	32 (89)	244 (97)	32 (89)	242 (96)	32 (89)
by patient	172 (68)	23 (64)	174 (69)	23 (64)	172 (68)	23 (64)
with assistance of caretaker	8 (3)	1 (3)	8 (3)	1 (3)	8 (3)	1 (3)
with assistance of clinician	63 (25)	8 (22)	62 (25)	8 (22)	62 (25)	8 (22)
Cycle 5, week 1, day 1						
Number of patients on study	208	24	208	24	208	24
Completed	198 (95)	21 (88)	199 (96)	21 (88)	197 (95)	21 (88)
by patient	144 (69)	16 (67)	144 (69)	16 (67)	142 (68)	16 (67)
with assistance of caretaker	7 (3)	0	8 (4)	0	8 (4)	0
with assistance of clinician	47 (23)	5 (21)	47 (23)	5 (21)	47 (23)	5 (21)
Cycle 6, week 1, day 1						
Number of patients on study	182	14	182	14	182	14
Completed	170 (93)	13 (93)	172 (95)	13 (93)	170 (93)	13 (93)
by patient	108 (59)	8 (57)	109 (60)	8 (57)	109 (60)	8 (57)
with assistance of caretaker	8 (4)	0	8 (4)	0	8 (4)	0
with assistance of clinician	54 (30)	5 (36)	55 (30)	5 (36)	53 (29)	5 (36)
End of treatment						
Number of patients on study	354	192	354	192	354	192
Completed	175 (49)	87 (45)	177 (50)	89 (46)	177 (50)	87 (45)
by patient	105 (30)	57 (30)	106 (30)	58 (30)	104 (29)	57 (30)
with assistance of caretaker	8 (2)	3 (2)	9 (3)	4 (2)	8 (2)	3 (2)
with assistance of clinician	62 (18)	27 (14)	62 (18)	27 (14)	65 (18)	27 (14)

BPI-SF=Brief Pain Inventory – Short Form. EQ-5D-5L=EuroQol 5-dimension 5-level. FACT-P=Functional Assessment of Cancer Therapy–Prostate. PSMA=prostate-specific membrane antigen. SoC=standard of care.
Completed is defined as having at least one response captured at the scheduled visit. Analyses in the 581 patients randomised after measures were implemented on or after March 5, 2019.

Table S3: Reasons for non-completion of patient-reported outcome questionnaires

	FACT-P		BPI-SF		EQ-5D-5L	
Study time-point	**^{177}Lu-PSMA-617 plus SoC (n = 385) n (%)**	**SoC alone (n = 196) n (%)**	**^{177}Lu-PSMA-617 plus SoC (n = 385) n (%)**	**SoC alone (n = 196) n (%)**	**^{177}Lu-PSMA-617 plus SoC (n = 385) n (%)**	**SoC alone (n = 196) n (%)**
Screening						
Number of patients	385	196	385	196	385	196
Not completed	18 (5)	10 (5)	16 (4)	10 (5)	18 (5)	10 (5)
Patient too ill	0	0	0	0	0	0
Inconvenient/time consuming	1 (1)	1 (1)	1 (1)	1 (1)	1 (1)	1 (1)
Administrative error	11 (3)	6 (3)	11 (3)	6 (3)	11 (3)	6 (3)
Other	6 (2)	3 (2)	4 (1)	3 (2)	6 (2)	3 (2)
Cycle 1, week 1, day 1						
Number of patients still on study	369	164	369	164	369	164
Not completed	20 (5)	18 (11)	12 (3)	13 (8)	20 (5)	17 (10)
Patient too ill	0	0	0	0	0	0
Inconvenient/time consuming	0	1 (1)	0	1 (1)	0	1 (1)
Administrative error	8 (2)	8 (5)	6 (2)	6 (4)	8 (2)	7 (4)
Other	12 (3)	9 (5)	6 (2)	6 (4)	12 (3)	9 (5)
Cycle 2, week 1, day 1						
Number of patients still on study	349	115	349	115	349	115
Not completed	14 (4)	13 (11)	10 (3)	13 (11)	14 (4)	13 (11)
Patient too ill	0	3 (3)	0	3 (3)	0	3 (3)
Inconvenient/time consuming	1 (1)	1 (1)	1 (1)	1 (1)	1 (1)	1 (1)
Administrative error	8 (2)	7 (6)	6 (2)	7 (1)	8 (2)	7 (6)
Other	5 (1)	2 (2)	3 (1)	2 (2)	5 (1)	2 (2)
Cycle 3, week 1, day 1						
Number of patients still on study	308	61	308	61	308	61
Not completed	16 (5)	6 (10)	11 (4)	7 (11)	12 (4)	6 (10)
Patient too ill	1 (1)	0	1 (1)	0	1 (1)	0
Inconvenient/time consuming	1 (1)	0	1 (1)	0	1 (1)	0
Administrative error	5 (2)	4 (7)	4 (1)	4 (7)	4 (1)	4 (7)
Other	9 (3)	2 (3)	5 (2)	3 (5)	6 (2)	2 (3)
Cycle 4, week 1, day 1						
Number of patients still on study	252	36	252	36	252	36
Not completed	9 (4)	4 (11)	8 (3)	4 (11)	10 (4)	4 (11)
Patient too ill	0	0	0	0	0	0
Inconvenient/time consuming	1 (1)	0	1 (1)	0	1 (1)	0
Administrative error	4 (2)	3 (8)	4 (2)	3 (8)	5 (2)	3 (8)
Other	4 (2)	1 (3)	3 (1)	1 (3)	4 (2)	1 (3)
Cycle 5, week 1, day 1						
Number of patients still on study	208	24	208	24	208	24
Not completed	10 (5)	3 (13)	9 (4)	3 (13)	11 (5)	3 (13)
Patient too ill	1 (1)	0	1 (1)	0	1 (1)	0
Inconvenient/time consuming	1 (1)	0	1 (1)	0	1 (1)	0
Administrative error	1 (1)	3 (13)	0	3 (13)	2 (1)	3 (13)
Other	7 (3)	0	7 (3)	0	7 (3)	0
Cycle 6, week 1, day 1						
Number of patients still on study	182	14	182	14	182	14
Not completed	12 (7)	1 (7)	10 (5)	1 (7)	12 (7)	1 (7)
Patient too ill	0	0	0	0	0	0
Inconvenient/time consuming	2 (1)	0	2 (1)	0	2 (1)	0
Administrative error	4 (2)	1 (7)	4 (2)	1 (7)	4 (2)	1 (7)
Other	6 (3)	0	4 (2)	0	6 (3)	0

BPI-SF=Brief Pain Inventory – Short Form. EQ-5D-5L=EuroQol 5-dimension 5-level. FACT-P=Functional Assessment of Cancer Therapy–Prostate. PSMA=prostate-specific membrane antigen. SoC=standard of care.
Completed is defined as having at least one response captured at the scheduled visit. Analyses in the 581 patients randomised after measures were implemented on or after March 5, 2019.

Table S4: Patient-reported outcome scores and changes from baseline by visit during randomised treatment

Patient-reported outcomes scores	^{177}Lu-PSMA-617 plus SoC (n = 385)					SoC alone (n = 196)				
	n	Mean	SD	Median	IQR	n	Mean	SD	Median	IQR
FACT-P total score										
Baseline	352	108·4	20·35	109·4	94·3–124·0	186	110·1	19·94	109·7	97·0–124·7
Cycle 2, week 1, day 1	334	112·7	20·68	115·0	98·0–129·0	102	105·6	22·44	102·3	89·0–126·0
Change from baseline	304	3·6	16·63	4·6	-6·0–13·0	100	-7·2	17·85	-4·0	-18·0–5·0
Cycle 3, week 1, day 1	292	113·2	20·44	115·0	101·5–128·8	54	110·2	21·11	106·9	96·0–125·8
Change from baseline	269	3·8	17·48	3·0	-5·5–15·0	52	-2·6	14·00	-3·5	-11·0–3·9
Cycle 4, week 1, day 1	243	114·3	21·64	117·0	99·0–130·7	32	110·5	21·30	109·3	93·0–130·5
Change from baseline	222	5·4	15·93	6·1	-5·0–16·0	32	-1·3	18·40	-2·5	-12·0–7·5
Cycle 5, week 1, day 1	198	114·1	21·34	117·0	101·0–130·0	21	106·3	26·25	102·0	87·3–123·3
Change from baseline	181	4·0	16·24	3·8	-5·0–14·0	21	-5·9	27·83	-5·0	-24·0–8·0
Cycle 6, week 1, day 1	169	114·3	22·26	119·0	98·0–131·0	13	114·5	23·98	122·0	98·0–128·0
Change from baseline	158	4·1	15·22	4·8	-5·0–13·0	13	-5·0	26·87	3·0	-16·0–16·0
End of treatment	175	101·7	24·14	103·2	82·5–121·0	87	101·8	25·29	103·0	86·0–124·0
Change from baseline	161	-9·4	21·64	-7·0	-20·0–4·0	86	-10·4	18·59	-7·5	-23·0–1·0
BPI-SF pain intensity scale										
Baseline	354	2·57	2·223	2·25	0·25–4·50	186	2·65	2·127	2·38	0·75–4·0
Cycle 2, week 1, day 1	339	1·87	1·951	1·50	0·00–3·25	102	2·74	2·369	2·50	0·50–4·25
Change from baseline	311	-0·59	2·037	0·00	-1·75–0·50	100	0·21	2·404	0·00	-1·00–1·63
Cycle 3, week 1, day 1	297	1·78	1·855	1·25	0·00–2·75	54	2·32	2·412	1·63	0·50–3·50
Change from baseline	274	-0·62	1·924	-0·25	-1·50–0·25	52	0·02	2·033	0·00	-1·13–1·50
Cycle 4, week 1, day 1	244	1·94	2·089	1·25	0·00–3·50	32	2·66	2·210	2·38	0·63–4·25
Change from baseline	223	-0·42	2·017	-0·25	-1·50–0·25	32	0·26	2·383	0·00	-1·00–1·50
Cycle 5, week 1, day 1	199	1·81	1·966	1·25	0·00–3·00	21	3·12	3·015	2·25	0·00–5·50
Change from baseline	183	-0·49	1·957	-0·25	-1·50–0·25	21	0·55	3·144	0·00	-1·00–1·75
Cycle 6, week 1, day 1	172	1·95	2·207	1·00	0·00–3·25	13	2·00	1·958	2·25	0·00–3·00
Change from baseline	162	-0·41	1·897	-0·25	-1·50–0·50	13	0·10	2·740	-0·75	-1·75–1·25
End of treatment	177	2·88	2·386	2·75	0·75–5·00	89	3·01	2·728	2·50	0·75–4·50
Change from baseline	165	0·46	2·415	0·00	-1·00–1·75	88	0·50	2·405	0·00	-1·00–2·00
EQ-5D-5L utility score										

Baseline	351	0·711	0·197	0·735	0·60–0·84	186	0·721	0·203	0·735	0·63–0·84
Cycle 2, week 1, day 1	334	0·742	0·209	0·749	0·65–0·88	102	0·648	0·256	0·675	0·57–0·84
Change from baseline	304	0·022	0·177	0·000	-0·06–0·12	100	-0·090	0·249	-0·015	-0·16–0·02
Cycle 3, week 1, day 1	296	0·749	0·197	0·767	0·65–0·88	55	0·715	0·201	0·735	0·62–0·84
Change from baseline	273	0·030	0·188	0·023	-0·06–0·12	53	-0·033	0·142	0·000	-0·10–0·04
Cycle 4, week 1, day 1	242	0·748	0·207	0·767	0·64–0·88	32	0·660	0·276	0·675	0·56–0·84
Change from baseline	221	0·029	0·177	0·007	-0·05–0·13	32	-0·082	0·238	-0·015	-0·14–0·06
Cycle 5, week 1, day 1	197	0·758	0·198	0·767	0·65–0·88	21	0·673	0·263	0·636	0·58–0·84
Change from baseline	180	0·040	0·168	0·020	-0·05–0·13	21	-0·067	0·286	-0·012	-0·12–0·05
Cycle 6, week 1, day 1	170	0·749	0·203	0·768	0·63–0·88	13	0·777	0·173	0·767	0·65–1·00
Change from baseline	159	0·034	0·180	0·012	-0·04–0·14	13	-0·011	0·178	0·000	-0·11–0·10
End of treatment	177	0·644	0·244	0·648	0·53–0·84	86	0·646	0·258	0·679	0·55–0·84
Change from baseline	163	-0·094	0·227	-0·091	-0·20–0·06	85	-0·090	0·212	-0·058	-0·21–0·01

BPI-SF=Brief Pain Inventory – Short Form. FACT-P=Functional Assessment of Cancer Therapy–Prostate. IQR=interquartile range. PSMA=prostate-specific membrane antigen. SoC=standard of care.

n for each visit is the number of patients with non-missing score at that visit.
n for change from baseline is the number of patients with non-missing score both at baseline and at that post-baseline visit.
Analyses in the 581 patients randomised after measures were implemented on or after March 5, 2019.

Table S5: Linear mixed effects model analysis on treatment differences in changes from baseline in patient-reported outcome scores

Patient reported outcomes scores	Treatment difference ^{177}Lu-PSMA-617 plus SoC – SoC alone LS mean (95% CI)
FACT-P total score	
Cycle 2, Week 1, Day 1	9·6 (6·0, 13·1)
Cycle 3, Week 1, Day 1	5·9 (1·6, 10·1)
Cycle 4, Week 1, Day 1	8·3 (3·5, 13·2)
Cycle 5, Week 1, Day 1	11 (4·6, 17·3)
Cycle 6, Week 1, Day 1	7·8 (0·6, 15·1)
Cycle 7, Week 1, Day 1	8·4 (0·4, 16·4)
End of Treatment	1·6 (-3·4, 6·6)
FACT-P physical well-being score	
Cycle 2, Week 1, Day 1	2·9 (1·9, 3·8)
Cycle 3, Week 1, Day 1	2·6 (1·4, 3·7)
Cycle 4, Week 1, Day 1	2·0 (0·6, 3·4)
Cycle 5, Week 1, Day 1	4·5 (2·7, 6·3)
Cycle 6, Week 1, Day 1	2·3 (0·3, 4·3)
Cycle 7, Week 1, Day 1	2·3 (0·1, 4·4)
End of Treatment	0·3 (-1·1, 1·7)
FACT-P social/family well-being score	
Cycle 2, Week 1, Day 1	0·1 (-0·9, 1·2)
Cycle 3, Week 1, Day 1	-1·1 (-2·2, 0·1)
Cycle 4, Week 1, Day 1	-0·9 (-2·2, 0·4)
Cycle 5, Week 1, Day 1	0·5 (-1·4, 2·3)
Cycle 6, Week 1, Day 1	0·3 (-1·7, 2·3)
Cycle 7, Week 1, Day 1	1·3 (-0·9, 3·5)
End of Treatment	0·5 (-0·7, 1·6)
FACT-P emotional well-being score	
Cycle 2, Week 1, Day 1	1·7 (1·0, 2·5)
Cycle 3, Week 1, Day 1	1·8 (1·0, 2·7)
Cycle 4, Week 1, Day 1	0·9 (-0·1, 2·0)
Cycle 5, Week 1, Day 1	2·0 (0·6, 3·4)
Cycle 6, Week 1, Day 1	1·4 (-0·5, 3·3)
Cycle 7, Week 1, Day 1	1·3 (-0·6, 3·3)
End of Treatment	-0·4 (-1·5, 0·8)
FACT-P functional well-being score	
Cycle 2, Week 1, Day 1	1·6 (0·5, 2·7)
Cycle 3, Week 1, Day 1	0·7 (-0·6, 2·1)
Cycle 4, Week 1, Day 1	2·0 (0·5, 3·5)
Cycle 5, Week 1, Day 1	1·0 (-0·9, 2·9)
Cycle 6, Week 1, Day 1	1·5 (-0·7, 3·7)
Cycle 7, Week 1, Day 1	3·5 (1·0, 6·0)
End of Treatment	0·3 (-1·1, 1·7)
FACT-P prostate cancer subscale score	

Cycle 2, Week 1, Day 1	2·8 (1·6, 4·0)
Cycle 3, Week 1, Day 1	1·1 (-0·4, 2·6)
Cycle 4, Week 1, Day 1	3·1 (1·2, 4·9)
Cycle 5, Week 1, Day 1	2·4 (0·3, 4·6)
Cycle 6, Week 1, Day 1	2·4 (-0·2, 4·9)
Cycle 7, Week 1, Day 1	-0·2 (-3·0, 2·6)
End of Treatment	0·6 (-1·3, 2·4)
FACT-P PCS pain-related subscale score	
Cycle 2, Week 1, Day 1	2·2 (1·4, 3·0)
Cycle 3, Week 1, Day 1	0·8 (-0·1, 1·8)
Cycle 4, Week 1, Day 1	1·5 (0·4, 2·7)
Cycle 5, Week 1, Day 1	1·6 (0·2, 3·0)
Cycle 6, Week 1, Day 1	1·0 (-0·7, 2·7)
Cycle 7, Week 1, Day 1	1·3 (-0·5, 3·0)
End of Treatment	0·1 (-0·9, 1·2)
FACT advanced prostate symptom index-8 (FAPSI-8) score	
Cycle 2, Week 1, Day 1	3·1 (2·1, 4·1)
Cycle 3, Week 1, Day 1	1·7 (0·5, 2·9)
Cycle 4, Week 1, Day 1	1·9 (0·5, 3·3)
Cycle 5, Week 1, Day 1	2·9 (1·1, 4·7)
Cycle 6, Week 1, Day 1	1·1 (-1·0, 3·2)
Cycle 7, Week 1, Day 1	1·1 (-1·0, 3·2)
End of Treatment	0·1 (-1·2, 1·4)
FACT-P trial outcome index score	
Cycle 2, Week 1, Day 1	7·5 (4·8, 10·2)
Cycle 3, Week 1, Day 1	4·9 (1·6, 8·1)
Cycle 4, Week 1, Day 1	7·8 (4·0, 11·6)
Cycle 5, Week 1, Day 1	8·4 (3·7, 13·2)
Cycle 6, Week 1, Day 1	5·9 (0·4, 11·4)
Cycle 7, Week 1, Day 1	5·8 (-0·3, 11·8)
End of Treatment	1·5 (-2·5, 5·4)
FACT-G total score	
Cycle 2, Week 1, Day 1	6·6 (3·9, 9·2)
Cycle 3, Week 1, Day 1	4·4 (1·2, 7·6)
Cycle 4, Week 1, Day 1	5·0 (1·4, 8·6)
Cycle 5, Week 1, Day 1	8·3 (3·5, 13·2)
Cycle 6, Week 1, Day 1	5·5 (0·0, 11·0)
Cycle 7, Week 1, Day 1	8·9 (2·9, 15·0)
End of Treatment	0·8 (-2·8, 4·4)
BPI-SF pain intensity scale	
Cycle 2, Week 1, Day 1	-0·82 (-1·22, -0·42)
Cycle 3, Week 1, Day 1	-0·58 (-1·03, -0·13)
Cycle 4, Week 1, Day 1	-0·64 (-1·22, -0·06)
Cycle 5, Week 1, Day 1	-1·00 (-1·68, -0·33)
Cycle 6, Week 1, Day 1	-0·31 (-1·14, 0·52)
Cycle 7, Week 1, Day 1	-0·32 (-1·15, 0·52)
End of Treatment	-0·11 (-0·65, 0·43)
BPI-SF pain interference scale	
Cycle 2, Week 1, Day 1	-1·03 (-1·47, -0·59)
Cycle 3, Week 1, Day 1	-0·25 (-0·80, 0·29)

Cycle 4, Week 1, Day 1	-0·88 (-1·53, -0·23)
Cycle 5, Week 1, Day 1	-1·16 (-1·93, -0·39)
Cycle 6, Week 1, Day 1	-0·83 (-1·76, 0·09)
Cycle 7, Week 1, Day 1	-0·48 (-1·38, 0·43)
End of Treatment	0·27 (-0·33, 0·87)
BPI-SF worst pain intensity scale	
Cycle 2, Week 1, Day 1	-1·16 (-1·71, -0·61)
Cycle 3, Week 1, Day 1	-0·74 (-1·38, -0·09)
Cycle 4, Week 1, Day 1	-0·71 (-1·59, 0·16)
Cycle 5, Week 1, Day 1	-1·05 (-1·97, -0·12)
Cycle 6, Week 1, Day 1	-0·63 (-1·80, 0·53)
Cycle 7, Week 1, Day 1	-0·82 (-1·94, 0·30)
End of Treatment	0·11 (-0·63, 0·84)
EQ-5D-5L utility score	
Cycle 2, Week 1, Day 1	0·1025 (0·0617, 0·1433)
Cycle 3, Week 1, Day 1	0·0505 (0·0043, 0·0966)
Cycle 4, Week 1, Day 1	0·0984 (0·0420, 0·1547)
Cycle 5, Week 1, Day 1	0·0956 (0·0291, 0·1621)
Cycle 6, Week 1, Day 1	0·0419 (-0·0337, 0·1174)
Cycle 7, Week 1, Day 1	0·0480 (-0·0323, 0·1283)
End of Treatment	0·0162 (-0·0370, 0·0695)

BPI-SF=Brief Pain Inventory – Short Form. CI=confidence interval. EQ-5D-5L=EuroQol 5-dimension 5-level. FACT-P=Functional Assessment of Cancer Therapy–Prostate. LS=least squares. PCS=prostate cancer subscale. PSMA=prostate-specific membrane antigen. SEM = standard error of mean. SoC=standard of care. Treatment difference estimated using linear mixed effects model analysis with the following factors and covariates: treatment, stratification factors, time (scheduled visit as a categorical variable), treatment by time interaction, baseline by time interaction, and baseline score. An unstructured correlation matrix was used to model the correlation within patients.

Table S6: Statistical analysis of time to worsening in BPI-SF scales defined as the time from randomisation to an increase of at least 2 points from baseline

Domain	*Post hoc* analyses ^{177}Lu-PSMA-617 plus SoC (n = 385)		SoC alone (n = 196)
BPI-SF pain intensity			
Time to worsening, n (%)			
Worsening relative to baseline	106 (28)		54 (28)
Censored[a]	279 (72)		142 (72)
Kaplan-Meier estimates (months)			
Median time to event, months (95% CI)	20·3 (14·9, NE)		6·4 (3·5, NE)
Hazard ratio		0·44	
95% CI		(0·31, 0·62)	
Two-sided p value		< 0·001	
BPI-SF pain interference			
Time to worsening, n (%)			
Worsening relative to baseline	136 (35)		56 (29)
Censored[a]	249 (65)		140 (71)
Kaplan-Meier estimates (months)			
Median time to event, months (95% CI)	14·9 (11·5, NE)		NE (4·4, NE)
Hazard ratio		0·53	
95% CI		(0·38, 0·73)	
Two-sided p value		< 0·001	
BPI-SF worst pain intensity			
Time to worsening, n (%)			
Worsening relative to baseline	156 (41)		75 (38)
Censored[a]	229 (59)		121 (62)
Kaplan-Meier estimates (months)			
Median time to event, months (95% CI)	12·1 (7·9, 15·1)		2·9 (1·9, 4·3)
Hazard ratio		0·49	
95% CI		(0·36, 0·65)	
Two-sided p value		< 0·001	

BPI-SF=Brief Pain Inventory–Short Form. CI=confidence interval.
[a]Patients without an event were censored at the last quality of life assessment. All p values are nominal, descriptive, and non-inferential. Analyses in the 581 patients randomised after measures were implemented on or after March 5, 2019.

Table S7: Summary of number of events and reason for censoring patients in time to worsening in BPI-SF scales in prespecified and post hoc analyses

Events and reason for censoring, n (%)	Prespecified analyses		*Post hoc* analyses	
	^{177}Lu-PSMA-617 plus SoC (n = 385)	SoC alone (n = 196)	^{177}Lu-PSMA-617 plus SoC (n = 385)	SoC alone (n = 196)
BPI-SF pain intensity				
Number of events				
Worsening relative to baseline	140 (36)	73 (37)	144 (37)	76 (39)
≥ 30% and ≥ 2 points	38 (10)	29 (15)	39 (10)	32 (16)
≥ 30% only	61 (16)	29 (15)	62 (16)	29 (15)
≥ 2 points only	41 (11)	15 (8)	43 (11)	15 (8)
Clinical progression	122 (32)	51 (26)		
Deaths	66 (17)	42 (21)		
Number of patients censored	57 (15)	30 (15)	241 (63)	120 (61)
Reason for censoring				
Censored at last BPI-SF assessment	57 (15)	17 (9)	223 (58)	85 (43)
No evaluable data (censored at randomisation)	0	13 (7)	18 (5)	35 (18)
BPI-SF pain interference				
Number of events				
Worsening relative to baseline	171 (44)	69 (35)	176 (46)	72 (37)
≥ 30% and ≥ 2 points	66 (17)	30 (15)	68 (18)	30 (15)
≥ 30% only	74 (19)	19 (10)	75 (19)	22 (11)
≥ 2 points only	31 (8)	20 (10)	33 (9)	20 (10)
Clinical progression	99 (26)	57 (29)		
Deaths	60 (16)	40 (20)		
Number of patients censored	55 (14)	30 (15)	209 (54)	124 (63)
Reason for censoring				
Censored at last BPI-SF assessment	55 (14)	17 (9)	191 (50)	89 (45)
No evaluable data (censored at randomisation)	0	13 (7)	18 (5)	35 (18)
BPI-SF worst pain intensity				
Number of events				
Worsening relative to baseline	167 (43)	81 (41)	170 (44)	83 (42)
≥ 30% and ≥ 2 points	69 (18)	38 (19)	72 (19)	39 (20)
≥ 30% only	29 (8)	15 (8)	29 (8)	16 (8)
≥ 2 points only	69 (18)	28 (14)	69 (18)	28 (14)
Clinical progression	109 (28)	49 (25)		
Deaths	56 (15)	39 (20)		
Number of patients censored	53 (14)	27 (14)	215 (56)	113 (58)
Reason for censoring				
Censored at last BPI-SF assessment	53 (14)	14 (7)	197 (51)	78 (40)
No evaluable data (censored at randomisation)	0	13 (7)	18 (5)	35 (18)

BPI-SF=Brief Pain Inventory–Short Form. SoC=standard of care.
Analyses in the 581 patients randomised after measures were implemented on or after March 5, 2019.

Table S8: Time to improvement after worsening in BPI-SF pain intensity scale

	^{177}Lu-PSMA-617 plus SoC (n = 385)	**SoC alone (n = 196)**
Worsening relative to baseline, n (%)	144 (37)	76 (39)
Time to improvement, n (%)		
Events	74 (51)	21 (28)
Censored[a]	70 (49)	55 (72)
Kaplan-Meier estimates (months)		
Median time to improvement (95% CI)	2·8 (1·9, 4·2)	4·2 (2·8, 11·3)
Hazard ratio	0·69	
95% CI	0·54, 0·87	
Two-sided p value	0·002	

BPI-SF=Brief Pain Inventory–Short Form. CI=confidence interval.
[a]Patients without an event were censored at the last quality of life assessment. Time to improvement is defined as the time from worsening of pain Intensity scale to a pain Intensity scale ≤baseline. All p values are nominal, descriptive, and non-inferential. Prespecified analyses in the 581 patients randomised after measures were implemented on or after March 5, 2019.

Table S9: Time to worsening of 0·10 points or greater in EQ-5D-5L utility score

	^{177}Lu-PSMA-617 plus SoC (n = 385)	SoC alone (n = 196)
Time to worsening, clinical disease progression or death		
Median time to worsening, months (95% CI)	5·9 (4·7, 7·1)	2·0 (1·6, 2·4)
Hazard Ratio	0·51	
95% CI	(0·42, 0·62)	
p value	<·001	
Time to worsening only		
Median time to worsening, months (95% CI)	11·2 (7·6, 14·0)	2·3 (1·6, 4·2)
Hazard Ratio	0·47	
95% CI	(0·35, 0·62)	
p value	<·001	

EQ-5D-5L=EuroQol 5-dimension 5-level.

Time to worsening in EQ-5D-5L was defined as the time from randomisation to a decrease in utility score of at least 0·10 points from baseline. Median time to worsening estimated from Kaplan-Meier curves censored for event (symptomatic skeletal event or death). All p values are nominal, descriptive, and non-inferential. Analyses were performed in the 581 patients randomised after measures were implemented on or after March 5, 2019.

Table S10: EQ-VAS values by visit

Analysis visit	n	Mean	SD	Median	IQR
^{177}Lu-PSMA-617 plus SoC (n = 385)					
Cycle 1, week 1, day 1	351	68·3	18·62	70	50·0–81·0
Cycle 2, week 1, day 1	335	71·4	19·63	75	60·0–89·0
Cycle 3, week 1, day 1	295	71·3	19·98	76	60·0–86·0
Cycle 4, week 1, day 1	242	73·2	19·03	79	60·0–89·0
Cycle 5, week 1, day 1	197	74·1	17·87	79	62·0–89·0
Cycle 6, week 1, day 1	170	72·2	21·13	75	60·0–90·0
End of treatment	177	62·6	23·71	65	50·0–81·0
SoC alone (n = 196)					
Cycle 1, week 1, day 1	185	68·5	20·01	70	52·0–81·0
Cycle 2, week 1, day 1	102	65·1	22·28	70	50·0–81·0
Cycle 3, week 1, day 1	55	69·6	23·18	71	60·0–90·0
Cycle 4, week 1, day 1	32	71	19·62	72	56·5–89·5
Cycle 5, week 1, day 1	21	64·6	25·6	75	48·0–81·0
Cycle 6, week 1, day 1	13	76·2	16·02	76	60·0–91·0
End of treatment	87	60	24·09	60	40·0–80·0

EQ-VAS=EuroQol-Visual Analogue Scale. IQR=interquartile range. PSMA=prostate-specific membrane antigen. SoC=standard of care.
n for each visit is the number of patients with non-missing score at that visit.
Analyses in the 581 patients randomised after measures were implemented on or after March 5, 2019.

Table S11: Subgroup analysis of time to first symptomatic skeletal event without death by concurrent use of bone-targeted agents as part of standard of care

	^{177}Lu-PSMA-617 plus SoC (n = 385)	SoC alone (n = 196)
Bone-targeted agents: Yes		
SSE, n/N (%)	23/175 (13)	17/96 (18)
Median time to event, months (95% CI)	NE (19·1, NE)	NE (10·5, NE)
HR (95% CI)	0·34 (0·18, 0·67)	
Median follow-up time, months (95% CI)	8·3 (6·7, 9·0)	2·8 (2·3, 3·7)
Bone-targeted agents: No		
SSE, n/N (%)	37/210 (18)	17/100 (17)
Median time to event, months (95% CI)	NE (18·2, NE)	NE (4·1, NE)
HR (95% CI)	0·35 (0·19, 0·66)	
Median follow-up time, months (95% CI)	8·8 (7·0, 10·2)	2·1 (1·7, 2·3)
Overall		
SSE, n/N (%)	60/385 (16)	34/196 (17)
Median time to event, months (95% CI)	NE (19·1, NE)	NE (10·5, NE)
HR (95% CI)	0·36 (0·23, 0·57)	
Median follow-up time, months (95% CI)	8·4 (7·1, 9·1)	2·3 (2·2, 2·7)

ARPI=androgen receptor pathway inhibitor. CI=confidence interval. ECOG=Eastern Cooperative Oncology Group. HR=hazard ratio. N=total number of events. N=total number of patients. NE=not estimable. PSMA=prostate-specific membrane antigen. SoC=standard of care. SSE=symptomatic skeletal event.
Median time to event and median follow-up time estimated from Kaplan-Meier curves censored for event (SSE or death). HR of ^{177}Lu-PSMA-617 + SoC *vs* SoC only is based on stratified Cox Proportional Hazards model stratified for lactate dehydrogenase (≤260 IU/L *vs* >260 IU/L); presence of liver metastases (yes *vs* no); ECOG score (0 or 1 *vs* 2); and inclusion of ARPI in best supportive/SoC at time of randomisation (yes *vs* no). Analyses in the 581 patients randomised after measures were implemented on or after March 5, 2019.

Table S12: Worst post-baseline haematological and creatinine abnormalities based on common terminology criteria for adverse events (CTCAE) grades during randomised treatment

	^{177}Lu-PSMA-617 plus SoC (n = 529)		SoC alone (n = 205)	
	All grades n (%)	**Grades 3/4 n (%)**	**All grades n (%)**	**Grades 3/4 n (%)**
Haemoglobin - decreased	520 (98)	80 (15)	179 (87)	13 (6)
Lymphocytes - decreased	480 (91)	269 (51)	141 (69)	39 (19)
Leukocytes - decreased	307 (58)	36 (7)	54 (26)	4 (2)
Platelets - decreased	258 (49)	49 (9)	49 (24)	5 (2)
Neutrophils - decreased	149 (28)	23 (4)	20 (10)	2 (1)
Eosinophils - increased	37 (7)	0	18 (9)	0
Haemoglobin - increased	1 (1)	0	0	0
Lymphocytes - increased	2 (1)	2 (1)	2 (1)	0
Creatinine - increased	157 (30)	5 (1)	47 (23)	1 (1)

CTCAE=Common Terminology Criteria for Adverse Events. PSMA=prostate-specific membrane antigen. SoC=standard of care. "All grades" represents patients with any grade 1, 2, 3, or 4 post-baseline. Grades based on CTCAE version 5·0. Analyses in the 734 patients who received at least one dose of randomised treatment.

Table S13: Haemoglobin levels by visit during randomised treatment

	^{177}Lu-PSMA-617 plus SoC (n = 529)					SoC alone (n = 205)				
	n	**Mean**	**SD**	**Median**	**IQR**	**n**	**Mean**	**SD**	**Median**	**IQR**
Haemoglobin (g/L)										
Baseline	529	118·4	15·25	118·0	107·0–130·	205	114·7	15·56	115·0	103·0–127·0
Cycle 1, week 3, day 1	501	117·6	15·39	119·0	106·3–129·0	172	113·9	16·53	113·5	102·0–127·0
Cycle 1, week 5, day 1	493	117·0	15·58	118·0	106·0–129·0	147	113·3	16·83	112·0	100·0–127·0
Cycle 2, week 1, day 1	501	116·5	14·77	117·0	106·0–128·0	134	113·5	18·44	114·0	100·0–128·0
Cycle 2, week 3, day 1	475	116·0	15·30	117·0	104·0–127·0	104	113·8	18·43	114·7	97·3–127·6
Cycle 2, week 5, day 1	443	115·6	15·72	118·0	104·0–127·0	72	112·4	16·62	115·0	100·0–125·5
Cycle 3, week 1, day 1	445	115·9	14·92	117·0	105·0–127·0	71	115·9	16·05	117·0	105·0–128·0
Cycle 3, week 3, day 1	411	115·9	15·24	117·0	106·0–127·0	55	114·0	16·52	114·0	102·0–128·0
Cycle 3, week 5, day 1	394	115·1	15·72	116·0	105·0–128·0	47	113·9	16·12	114·0	104·0–125·0
Cycle 4, week 1, day 1	368	115·8	14·88	116·0	104·0–128·0	44	113·3	15·34	111·6	103·0–127·0
Cycle 4, week 3, day 1	323	116·2	15·32	117·0	106·0–128·0	32	112·0	16·53	110·0	98·1–126·0
Cycle 4, week 5, day 1	307	114·8	16·27	116·0	104·0–127·0	34	111·6	17·11	112·0	102·0–125·0
Cycle 5, week 1, day 1	299	115·5	15·13	117·0	104·0–127·3	30	112·8	17·11	114·0	99·0–128·0
Cycle 5, week 3, day 1	257	115·3	14·66	117·0	105·0–126·0	18	113·0	16·49	114·7	104·0–122·0
Cycle 5, week 5, day 1	253	115·2	14·70	117·0	106·0–127·	16	112·9	20·63	112·0	98·0–132·0
Cycle 6, week 1, day 1	254	114·3	15·18	115·0	104·7–125·0	19	116·9	14·96	118·0	108·0–129·0
Cycle 6, week 3, day 1	205	114·7	14·30	115·0	105·0–125·0	13	116·8	16·93	113·0	110·0–132·0
Cycle 6, week 5, day 1	200	113·5	14·56	113·0	103·5–124·0	14	117·4	16·68	116·5	108·0–132·0
End of treatment	326	104·1	17·31	105·0	92·0–116·0	132	112·3	18·72	111·1	98·0–126·5

IQR=interquartile range. PSMA=prostate-specific membrane antigen. SoC=standard of care.
Analyses in the 734 patients who received at least one dose of randomised treatment.

Table S14: Shift table for change in haemoglobin levels based on CTCAE grades during randomised treatment

Treatment	Baseline		Worst post-baseline value					
	Grade	n (%)	Grade 0 n (%)	Grade 1 n (%)	Grade 2 n (%)	Grade 3 n (%)	Grade 4 n (%)	Missing n (%)
^{177}Lu-PSMA-617 plus SoC (n = 529)	Grade 0	119 (22)	9 (8)	93 (78)	14 (12)	3 (3)	0	0
	Grade 1	350 (66)	0	159 (45)	149 (43)	42 (12)	0	0
	Grade 2	59 (11)	0	2 (3)	23 (39)	34 (58)	0	0
	Grade 3	1 (1)	0	0	0	1 (100)	0	0
	Grade 4	0	0	0	0	0	0	0
	Missing	0	0	0	0	0	0	0
	Total	529 (100)	9 (2)	254 (48)	186 (35)	80 (15)	0	0
SoC alone (n = 205)	Grade 0	38 (19)	17 (45)	19 (50)	1 (3)	0	0	1 (3)
	Grade 1	131 (64)	2 (2)	83 (63)	35 (27)	7 (5)	0	4 (3)
	Grade 2	36 (18)	0	0	28 (78)	6 (17)	0	2 (6)
	Grade 3	0	0	0	0	0	0	0
	Grade 4	0	0	0	0	0	0	0
	Missing	0	0	0	0	0	0	0
	Total	205 (100)	19 (9)	102 (50)	64 (31)	13 (6)	0	7 (3)

CTCAE=Common Terminology Criteria for Adverse Events. PSMA=prostate-specific membrane antigen. SoC=standard of care.
Baseline percentage is based on the number patients who received at least one dose of randomised treatment. Worst post-baseline percentage is based on the number of patients at baseline. Grades based on CTCAE version 5·0. Analyses in the 734 patients who received at least one dose of randomised treatment.

Table S15: Incidence, severity, and outcome of adverse events in the myelosuppression safety topic of interest during randomised treatment

Events and censoring	^{177}Lu-PSMA-617 plus SoC (n = 529) n (%)	SoC alone (n = 205) n (%)
Patients with at least one event	251 (47)	36 (18)
Anaemia	168 (32)	27 (13)
Thrombocytopenia	91 (17)	9 (4)
Lymphopenia	75 (14)	8 (4)
Leukopenia	66 (12)	4 (2)
Neutropenia	45 (9)	3 (1)
Pancytopenia	8 (2)	0
Febrile neutropenia	2 (1)	0
Bicytopenia	1 (1)	0
Bone marrow failure	1 (1)	0
Normocytic anaemia	1 (1)	0
Maximum CTCAE grade		
Grade 3	104 (20)	14 (7)
Grade 4	17 (3)	0
Grade 5	3 (1)	0
TRAEs	215 (41)	10 (5)
SAEs	27 (5)	1 (1)
Action taken with ^{177}Lu-PSMA-617		
Drug withdrawn	37 (7)	NA
Dose reduced	21 (4)	NA
Drug interrupted	50 (9)	NA
Dose not changed/NA/unknown	233 (44)	NA
Action taken with SoC		
Drug withdrawn	14 (3)	0
Dose reduced	0	0
Drug interrupted	12 (2)	0
Dose not changed/NA/unknown	246 (47)	36 (18)
AE outcome		
Recovered/resolved	123 (23)	17 (8)
Recovering/resolving	4 (1)	1 (1)
Not recovered/not resolved	188 (36)	26 (13)
Recovered/resolved with sequelae	3 (1)	0
Fatal	3 (1)	0
Unknown	3 (1)	0

CTCAE=Common Terminology Criteria for Adverse Events. PSMA=prostate-specific membrane antigen. SAE=serious adverse event. SoC=standard of care. TRAE=treatment-related adverse event.

Baseline percentage is based on the number patients who received at least one dose of randomised treatment. Worst post-baseline percentage is based on the number of patients at baseline. Grades based on CTCAE version 5·0. Analyses in the 734 patients who received at least one dose of randomised treatment

Study sites and enrollment

Country	City	Site name	Principal investigator	Patients (N = 831)
USA	Rochester	Mayo Clinic - Rochester	Ayse Kendi	54
USA	Omaha	Urology Cancer Center, PC	Luke Nordquist	46
USA	Detroit	Karmanos Cancer Institute - Detroit	Nitin Vaishampayan	30
France	Villejuif	Gustave Roussy Oncology Institute	Karim Fizazi	26
USA	Tampa	H. Lee Moffitt Cancer Center & Research Institute	Ghassan El-Haddad	26
Canada	Vancouver	BC Cancer - Vancouver	Kim Chi	25
USA	Las Vegas	Comprehensive Cancer Centers of Nevada - Twain Office	Oscar Goodman, Jr.	25
USA	New York	Memorial Sloan-Kettering Cancer Center	Michael Morris	25
USA	New Orleans	Tulane Medical Center, Tulane Cancer Center	Alton O. Sartor	24
USA	Los Angeles	University of California Los Angeles, Nuclear Medicine	Johannes Czernin	24
USA	Portland	Oregon Health and Science University, Nuclear Medicine	Jacqueline Vuky	21
France	Lyon	Leon Berard Center	Aude Flechon	20
USA	Houston	Excel Diagnostics & Nuclear Oncology Center	Ebrahim Delpassand	19
Netherlands	Nijmegen	Radboud University Medical Center (Radboudumc)	James Nagarajah	19
UK	Sutton	Institute of Cancer Research	Johann de Bono	19
USA	Aurora	University of Colorado Hospital	Bennett Chin	19
USA	Saint Louis	Washington University School of Medicine	Jeff Michalski	16
Denmark	Copenhagen	Rigshospitalet - University Hospital Copenhagen	Peter Petersen	15
USA	Louisville	Norton Cancer Institute	Chandler Park	15
USA	Tucson	University of Arizona Cancer Center	Edward Gelmann	15
USA	Dallas	UT Southwestern Medical Center	Orhan Oz	15
France	Clermont-Ferrand	Center Jean Perrin	Hakim Mahammedi	14
Sweden	Stockholm	Karolinska University Hospital	Enrique Castellanos	14
USA	Iowa City	University of Iowa Hospitals and Clinics	Yousef Zakharia	14
USA	Scottsdale	HonorHealth Research Institute	Michael Gordon	13
USA	Boston	Dana Farber Cancer Institute	Xiao Wei	12
USA	Palo Alto	Stanford Cancer Institute	Sandhya Srinivas	12
Belgium	Anderlecht	Jules Bordet Institute	Carlos Artigas Guix	11
Netherlands	Utrecht	UMC Utrecht	Bart de Keizer	11
USA	Indianapolis	Indiana University Melvin and Bren Simon Cancer Center	Nabil Adra	11
USA	New York	New York Presbyterian Hospital/Weill Cornell Medical Center	Scott Tagawa	11
USA	Durham	Duke University Medical Center, Duke Cancer Center	Andrew Armstrong	10
USA	East Brunswick	Regional Cancer Care Associates, Central Jersey Division	Bruno Fang	10
UK	Glasgow	Beatson West of Scotland Cancer Center	Robert Jones	8
USA	Chicago	Northwestern University	David VanderWeele	8
Denmark	Aalborg	Aalborg University Hospital, Oncology Department	Mette Moe Kempel	7
USA	Myrtle Beach	Carolina Urologic Research Center	Neal Shore	7
USA	San Francisco	UCSF Helen Diller Family Comprehensive Cancer Center	Vadim Koshkin	7
Canada	Quebec	CHU of Quebec - Laval University	Jean-Mathieu Beauregard	6
Canada	Montreal	CHUM - University Hospital of Montreal	Fred Saad	6
Canada	Toronto	Sunnybrook Research Institute, Odette Cancer Center	Urban Emmenegger	6
France	Toulouse	Institute Claudius Regaud, Toulouse Cancer Research Center	Frederic Courbon	6
Sweden	Umea	Norrlands University Hospital, Cancer Center	Anders Widmark	6

Sweden	Lund	Skane University Hospital - Barngatan, Clinical Trials Unit	Olof Stahl	6
USA	Charlottesville	Emily Couric Clinical Cancer Center	Robert Dreicer	6
USA	Albuquerque	New Mexico Cancer Center	Gregg Franklin	6
USA	Ann Arbor	University of Michigan Hospitals	Morand Piert	6
Netherlands	Nieuwegein	St. Antonius Hospital	Jules Lavalaye	5
UK	London	St Bartholomew's Hospital	Jonathan Shamash	5
USA	Kettering	Greater Dayton Cancer Center	Charles Bane	5
USA	Fort Wayne	Parkview Research Center	Brian Chang	5
Sweden	Uppsala	Uppsala University Hospital, Department of Oncology	Vasileios Kostaras	4
UK	London	Guy's Hospital	Deborah Enting	4
USA	Gettysburg	Pennsylvania Cancer Specialists & Research Institute	Satish Shah	4
USA	Philadelphia	Thomas Jefferson University Hospital	Jessie DiNome	4
USA	Ann Arbor	VA Ann Arbor Healthcare System	David Elliott	4
USA	Dallas	VA North Texas Health Care System, Nuclear Medicine Service	Irfan Farukhi	4
Belgium	Brussels	Saint Luc University Hospital	Renaud Lhommel	3
Belgium	Leuven	University Hospitals Leuven, Department of Nuclear Medicine	Benoit Beuselinck	3
Canada	Montreal	Jewish General Hospital	Cristiano Ferrario	3
France	Bordeaux	Bergonie Institute	Anne Cazeau	3
Netherlands	Amsterdam	The Netherlands Cancer Institute	Wouter Vogel	3
Sweden	Gothenburg	Sahlgrenska University Hospital, Department of Oncology	Jon Kindblom	3
UK	Bristol	Bristol Haematology and Oncology Center	Amit Bahl	3
UK	Guildford	Royal Surrey County Hospital	Chee Goh	3
UK	Southampton	University Hospital Southampton NHS Foundation Trust	Simon Crabb	3
USA	Towson	Chesapeake Urology Associates (CUA) P.A.	Ronald Tutrone, Jr.	3
USA	Saint Louis	Saint Louis University Hospital	Medhat Osman	3
USA	Seattle	Swedish Cancer Institute Research	Song Zhao	3
USA	Baltimore	University of Maryland Greenebaum Cancer Center	Arif Hussain	3
Canada	London	London Health Sciences Centre	David Laidley	2
Denmark	Aarhus	Aarhus University Hospital, Department of Oncology	Simon Buus	2
UK	London	University College London Hospitals NHS Foundation Trust	Mark Linch	2
USA	Boston	Beth Israel Deaconess Medical Center	Glenn Bubley	2
USA	Washington, DC	Washington DC VA Medical Center, Nuclear Medicine Service	Frank Liu	2
Canada	Ottawa	Ottawa Hospital, Cancer Center	Michael Ong	1
France	Paris	Tenon Hospital	Ahmed Khalil	1
Puerto Rico	Puerto Rico	VA Caribbean Healthcare System	Irma Molina-Vicenty	1
USA	Los Angeles	VA Greater Los Angeles Healthcare System	Gholam Reza Berenji	1
USA	Saint Louis	VA St. Louis Health Care System - John Cochran	Medhat Osman	1
USA	New Haven	Yale Cancer Center	Daniel Petrylak	1